



UNIVERSITY OF
WATERLOO

MTE 203 – Advanced Calculus

Midterm Exam Fall 2020 – Problem 4

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If $u(x, y) = x^2 f\left(\frac{y}{x}\right)$, where f is a differentiable function, use the chain rule to show that $u(x, y)$ satisfies,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$$

Solution:

Index: Chain Rule



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